University of Engineering & Technology

Lahore

INDUSTRIAL AUTOMATION

Dual-Zone Industrial Thermal Process Controller

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ABSTRACT

This project describes the design and development of a Dual-Zone Industrial Thermal Process Control System using a Programmable Logic Controller. This system is designed to control the temperature of two independent zones, Zone A and Zone B, using a closed-loop PID control system. The temperature is fed back through the analog inputs of the Programmable Logic Controller, which measures the temperature in engineering units through scaling instructions.
Every zone follows a prescribed industrial thermal profile, which includes the heating, soaking, cooling, and hold steps. Digital outputs drive the heaters and cooling fans, preventing both actuators from being engaged at the same time via internally prescribed PLC logic.
For its safe and reliable performance, this system is provided with an emergency stop function, protection against high temperatures, soak deviation warnings, and visual warnings from an LED with specified duty cycle. This project work is an example of the use of PLC program-implementation techniques such as state machine, timer, PID control, and analog signal processing, and safety-interlock functions. The program has been simulated to test its correctness under various conditions.

Introduction

Industrial automation has found numerous applications in the current manufacturing process, where accuracy, safety, reliability, and productivity are improved. Temperature control systems are widely found in industries such as chemical processing, heat treatment, food processing, and material testing. PLC-based control systems are widely preferred in such applications owing to their ruggedness and flexibility.
The Industrial Automation Lab is where students learn to create practical automation solutions using PLCs, sensors, actuators, and control algorithms. The purpose of this project will be to implement a two-zone thermal control system, which is practically equivalent to real-world heating and cooling applications. It employed analog temperature signals sourced from the trainer board, which were then processed in the PLC through the use of NORM_X and SCALE_X instructions to get the temperatures in degrees Celsius. Using the temperatures, the PID controllers controlled the heaters and the cooling fans with the desired deadband precision
A control scheme based on a state machine handles various stages involved in a thermal process. The code in this way is organized, readable, and easy to debug. The safety features in industrial processes, such as emergency stop, overtemperature shut down, or alarm handling, are incorporated to ensure conformance with industrial automation practices.
This project improves comprehension of:

- PLC programming using TIA Portal
- Analog signal processing
- PID Temperature Control
- State machine design
- Industrial safety and alarm systems

Input / Output Tags

Digital Inputs

Tag Name	Address	Description
PB_ESTOP	**%I0.0**	**Emergency Stop Push Button**
PB_RESET	**%I0.1**	**Alarm Reset Push Button**
SW1_HighAccuracy	**%I0.2**	**High Accuracy Mode Switch**
SW2_TestMode	**%I0.3**	**Over-Temperature Test Mode**
FirstScan	**%M1.0**	**PLC First Scan Bit**

Digital Outputs

Tag Name	Address	Description
Q_ZoneA_Heater	**%Q0.0**	**Zone-A Heater Output**
Q_ZoneA_Fan	**%Q0.1**	**Zone-A Fan Output**
Q_ZoneB_Heater	**%Q0.2**	**Zone-B Heater Output**
Q_ZoneB_Fan	**%Q0.3**	**Zone-B Fan Output**
Alarm_LED	**%Q0.4**	**Alarm Indication LED**

Internal Bool Flags

Tag Name	Address	Description
SystemFault	**%M100.0**	**System Fault Flag**
HighTempAlarm_A	**%M110.1**	**Zone-A High Temperature Alarm**
HighTempAlarm_B	**%M110.2**	**Zone-B High Temperature Alarm**
SoakAlarm_A	**%M100.3**	**Zone-A Soak Deviation Alarm**
PID_SET_HEATER	**%M100.4**	**Zone-A PID Heater Enable**
PID_SET_FAN	**%M100.5**	**Zone-A PID Fan Enable**
PID_SET_HEATER_B	**%M100.6**	**Zone-B PID Heater Enable**

	PID_SET_FAN_B	**%M100.7**	**Zone-B PID Fan Enable**

Analog Inputs

Tag Name	Address	Description
AI_ZoneA_Raw	**%IW64**	**Zone-A Temperature Sensor Raw Value**
AI_ZoneB_Raw	**%IW66**	**Zone-B Temperature Sensor Raw Value**

Analog Values

Tag Name	Address	Description
ZoneA_Temp	**%MD60**	**Zone-A Temperature (°C)**
ZoneB_Temp	**%MD70**	**Zone-B Temperature (°C)**

Setpoints and States

Tag Name	Address	Description
SP_ZoneA	**%MD0**	**Zone-A Temperature Setpoint**
SP_ZoneB	**%MD4**	**Zone-B Temperature Setpoint**
ZoneA_State	**%MD204**	**Zone-A Process State**
ZoneB_State	**%MD200**	**Zone-B Process State**

Control and PID parameters

Tag Name	Address	Description
Deadband	**%MD8**	**Temperature Deadband (2°C / 4°C)**
Soak_Result	**%MD16**	**Soak Timer Result**
Result_Sub	**%MD22**	**Zone-A Lower Threshold Calculation**

	Result_Add	**%MD26**	**Zone-A Upper Threshold Calculation**
	Result_Sub_B	**%MD30**	**Zone-B Lower Threshold Calculation**
	Result_Add_B	**%MD34**	**Zone-B Upper Threshold Calculation**

Zone A: State Description Of Thermal Process Control

State 0: Idle

At this state, the system is ready to start. All the outputs, like the heater, the fan, are OFF. At the beginning, the PLC will initialize the process.

State 1: Preheat

The temperature is raised to 35°C.
The heater goes ON to heat, and the fan is turned OFF.
Applying dead band control, whose value depends on the accuracy chosen in the switch.

States 2: Soak

The system maintains the temperature at 35°C for 3 minutes. The system uses PID.
If the temperature exceeds the permissible value ± 2°C for a period of more than 10 seconds, a soak alarm is generated.

State 4: Finish

The set point temperature is altered to 30°C, with PID control holding this temperature for 2 minutes. Once this process is done, the process can go back to idle state.
All safety conditions are dealt with in this section. If the Emergency Stop button is activated, the outputs will IMMEDIATELY go OFF, and the alarm LED will start blinking.

Fault State

This state becomes active in case of emergency stop or high-temperature situation.

All outputs are in the OFF condition, with the alarm LED flashing until the system resets.

High Temperature Protection

In case the temperature surpasses 45°C, the heater turns OFF automatically, and the fan is employed to cool the system till it achieves a stable temperature.

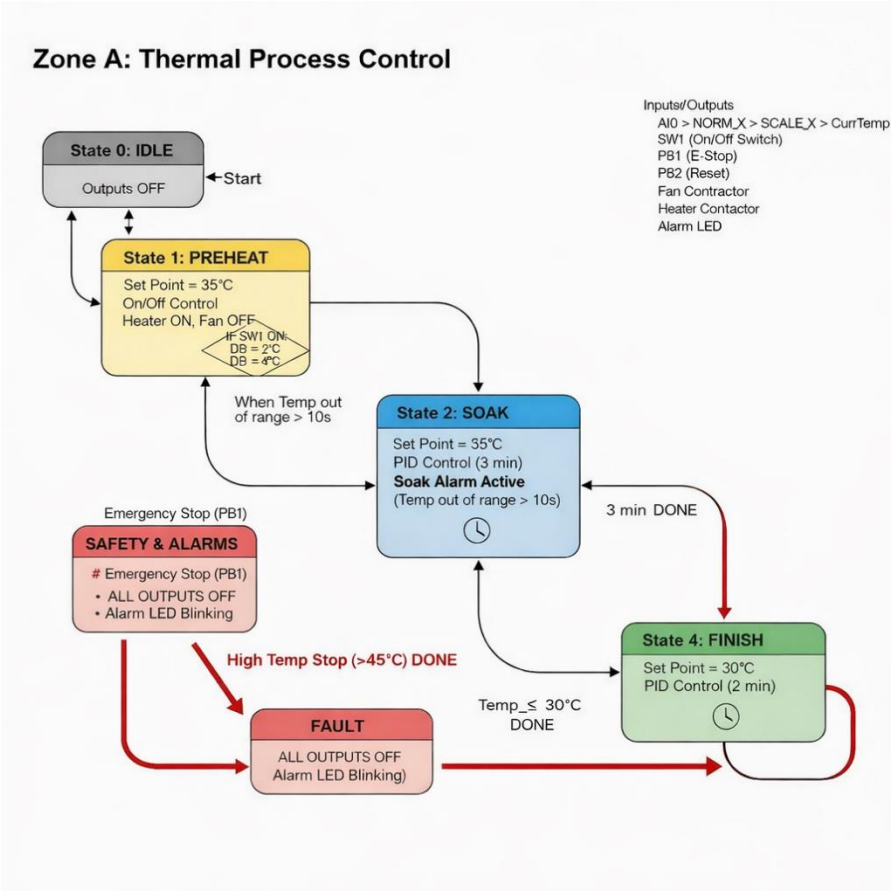


FIG 1: State Diagram of Zone A

Zone B: Thermal Process Control

State 0: Idle

The system is resting.
Nothing is ON.
It waits for the start or reset button.

State 1: Heat Up

The system heats the zone to **40°C**.
The heater turns ON to increase the temperature.
The fan works only when needed, depending on the switch.

State 2: Soak

The temperature is kept at **40°C** for **2 minutes**.
The PLC controls the heater and fan to keep the temperature steady.

State 3: Cool Down

The system cools the zone to **35°C**.
The heater is OFF and the fan stays ON until the temperature goes down.

Safety and Alarm

This part keeps the system safe.
If the emergency stop button is pressed, everything turns OFF and the alarm light starts blinking.

Fault State

This state comes when there is a problem.
All outputs are OFF.
The system waits until the reset button is pressed.

High Temperature Protection

If the temperature becomes too high, the heater is turned OFF.
The fan runs to cool the system until it becomes safe again.

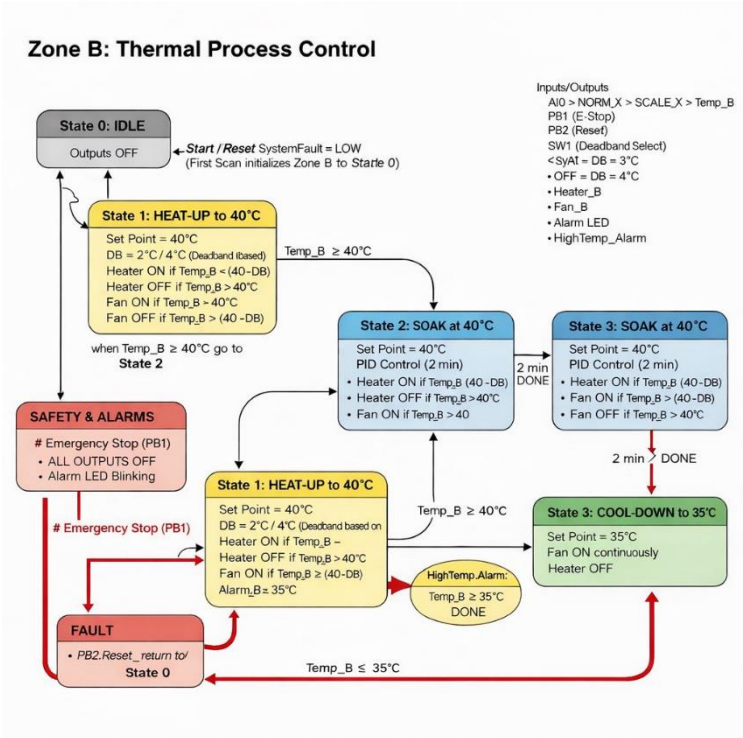
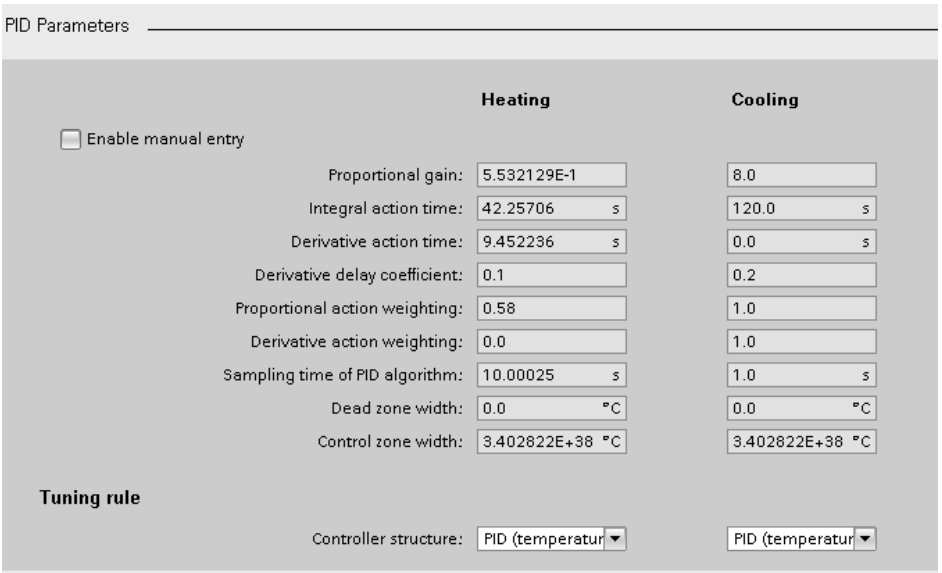


FIG 2: State Diagram of Zone A

Zone A – PID Control

Zone A uses PID to **keep the temperature steady**.The heater warms the system slowly and smoothly. The fan cools it when the temperature goes too high.These values help the temperature **not jump**, **not shake**, and **stay near the set point**.

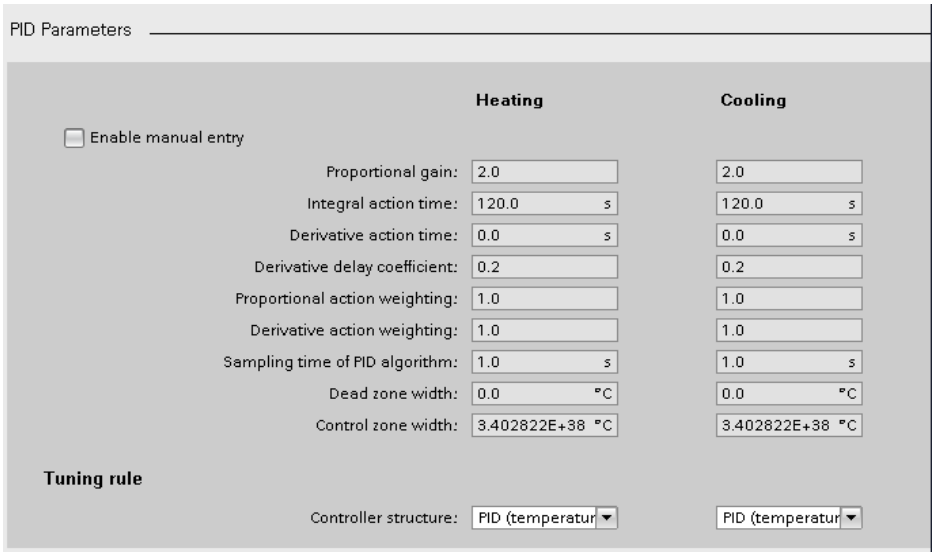


PID Parameters

	Heating	Cooling
Enable manual entry		
Proportional gain:	5.532129E-1	8.0
Integral action time:	42.25706 s	120.0 s
Derivative action time:	9.452236 s	0.0 s
Derivative delay coefficient:	0.1	0.2
Proportional action weighting:	0.58	1.0
Derivative action weighting:	0.0	1.0
Sampling time of PID algorithm:	10.00025 s	1.0 s
Dead zone width:	0.0 °C	0.0 °C
Control zone width:	3.402822E+38 °C	3.402822E+38 °C
Tuning rule		
Controller structure:	PID (temperatur	PID (temperatur

Zone B – PID Control

Zone B uses PID to **keep the temperature right**.The heater makes it warm, and the fan makes it cool.These values help the temperature **change slowly and stay safe**.

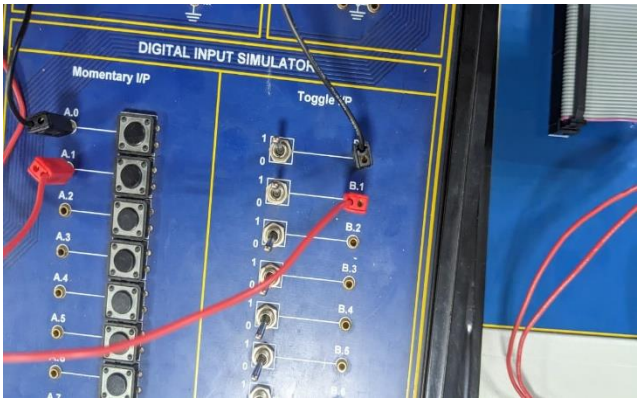


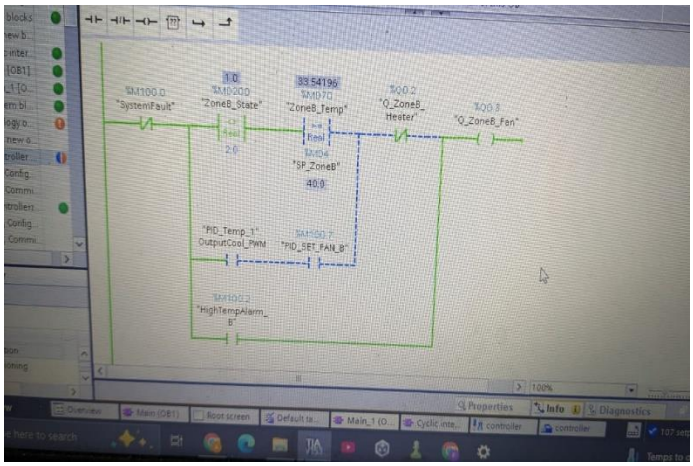
PID Parameters

	Heating	Cooling
Enable manual entry		
Proportional gain:	2.0	2.0
Integral action time:	120.0 s	120.0 s
Derivative action time:	0.0 s	0.0 s
Derivative delay coefficient:	0.2	0.2
Proportional action weighting:	1.0	1.0
Derivative action weighting:	1.0	1.0
Sampling time of PID algorithm:	1.0 s	1.0 s
Dead zone width:	0.0 °C	0.0 °C
Control zone width:	3.402822E+38 °C	3.402822E+38 °C
Tuning rule		
Controller structure:	PID (temperatur	PID (temperatur

Test Case 1: Test Switch Operation

Action:
The **Test switch** is pressed.
Expected Operation:
The **fan of both Zone A and Zone B turns ON**.
System Behavior:
The fans keep running until the temperature **reaches 25°C**.
Result:
When the temperature becomes **25°C**, the fans **turn OFF automatically**.



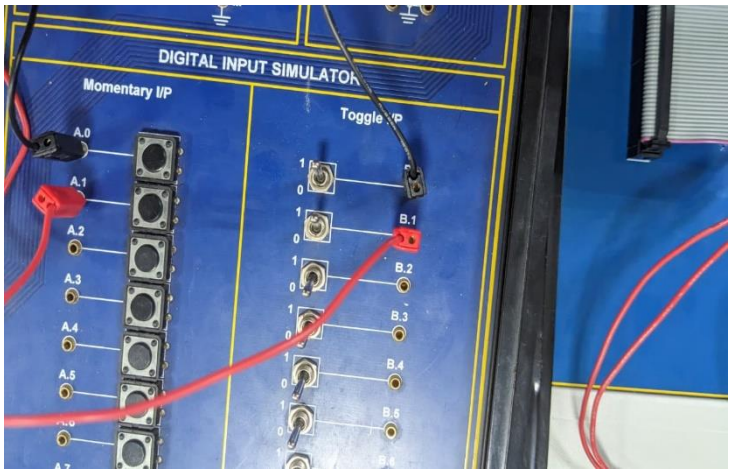
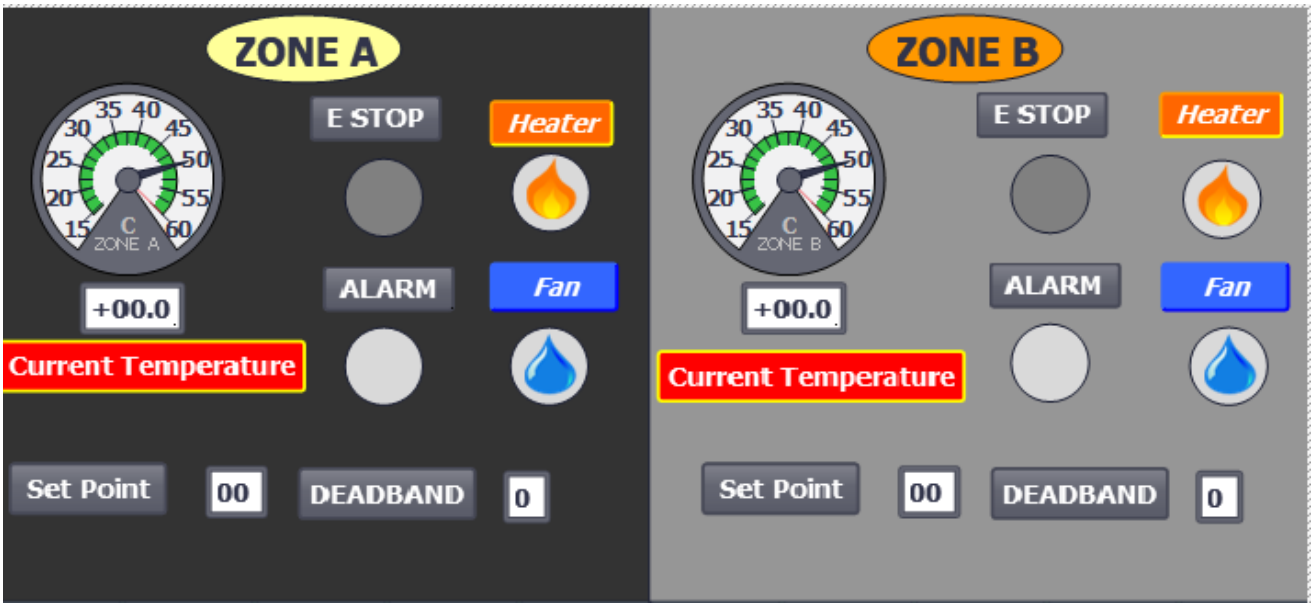


Test Case: Emergency Stop Button
Action:
The **Emergency Stop button** is pressed.

Expected Operation:
All outputs turn OFF immediately.
System Behavior:

- Heater OFF
- Fan OFF
- System stops at once

Result:
No output remains ON while the emergency button is active.



Final HMI